Ασκήσεις 1.5 (9/9)

Σωτήρης Γιαγιώτης
22983

1.1

$x_1(t) = \sin(10\pi t)$

$\sin(10\pi(t+T)) = \sin(10\pi t)$

$10\pi t + 10\pi T = 10\pi t + 2k\pi$

$10\pi T = 2k\pi \Rightarrow T = \frac{1}{5}k$

$k = 0, 1, 2$

Το σήμα είναι περιοδικό με θεμελιώδη περίοδο $T = \frac{1}{5}$ και συχνότητα $F = 5Hz$.

$x_S(t) = x_1(t) + x_3(t)$

$x_3(t) = \sin(31t)$

$x_3(t+T) = x_3(t)$

$\sin(31(t+T)) = \sin(31t)$

$31T = 2k\pi$

$T = \frac{2}{31}k\pi$

Το σήμα είναι περιοδικό με περίοδο $T_0 = \frac{2\pi}{31}$

$x_S(t) = x_1(t) + x_3(t)$

$x_1(t)$: περιοδικό με $T_0 = \frac{1}{5}$

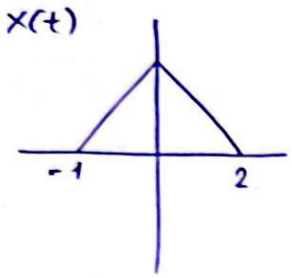
$x_3(t)$: περιοδικό με $T_0 = \frac{2\pi}{31}$

$\Rightarrow$ Άρα το $x_S(t)$ είναι περιοδικό με περίοδο ίση με $\text{ελπ}(T_{01}, T_{03})$, $T_0 = \frac{2\pi}{155}$

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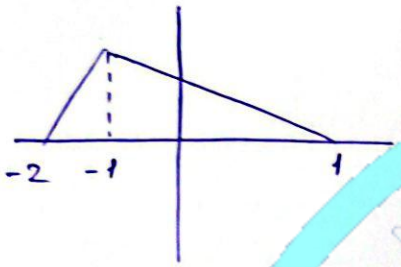
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1.2



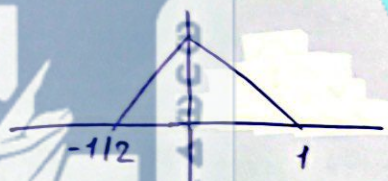
$$x(t)\begin{cases} 2t+2, & -1\le t<0 \\ 2-t, & 0\le t<2 \\ 0, & \text{αλλού} \end{cases}$$

$y_1(t) = x(t+1)$

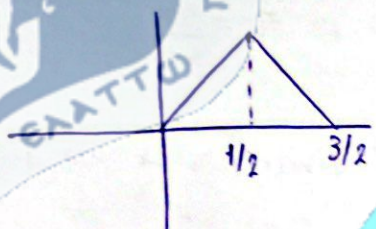


$$y_1(t)=\begin{cases} 2(t+1)+2, & -2\le t<-1 \\ 1-t, & -1\le t\le 1 \\ 0, & \text{αλλού} \end{cases}$$

$$y_2(t) = x(2t) = \begin{cases} 4t+2, & -\frac{1}{2}\le t<0 \\ 2-2t, & 0\le t<1 \\ 0, & \text{αλλού} \end{cases}$$



$$y_3(t) = x(2t-1) = \begin{cases} 4t, & 0\le t<\frac{1}{2} \\ 3-2t, & \frac{1}{2}\le t<\frac{3}{2} \\ 0, & \text{αλλού} \end{cases}$$

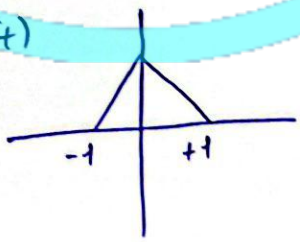


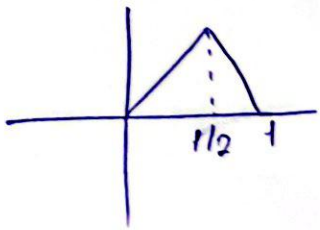
ΠΡΩΤΟΠΟΡΙΑ
ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

1.3

$x_2(t) = \Lambda(2t-1)$

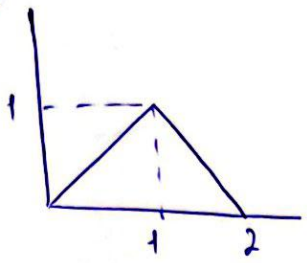
$$\Lambda(t) = \begin{cases} t+1, & -1\le t\le 0 \\ -t+1, & 0\le t\le 1 \\ 0, & \text{αλλού} \end{cases}$$





$$\Lambda(2t-1) = \begin{cases} 2t, & 0\le t\le \frac{1}{2} \\ -2t+2, & \frac{1}{2}\le t\le 1 \\ 0, & \text{αλλού} \end{cases}$$

Watermarkly



$$x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2-t, & 1 \le t \le 2 \\ 0, & \text{αλλού} \end{cases}$$

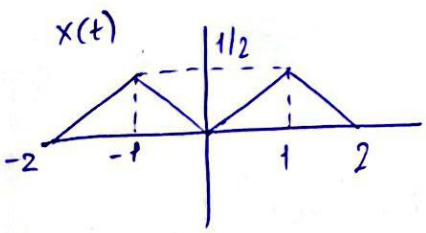
$$x_e(t) = \frac{1}{2}\left[x(t) + x(-t)\right] = \begin{cases} \frac{1}{2}\left[t+0\right] = \frac{t}{2}, & 0 \le t \le 1 \\ -\frac{t}{2}, & 1 \le t \le 2 \\ -\frac{t}{2}, & -1 \le t \le 0 \\ \frac{2+t}{2}, & -2 \le t \le -1 \\ 0, & \text{αλλού} \end{cases}$$

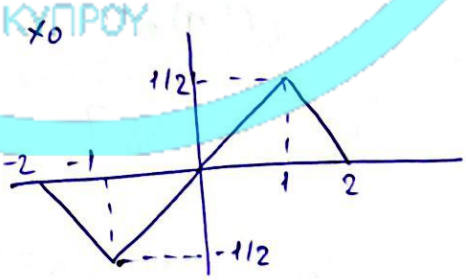
$$x(-t) = \begin{cases} -t, & -1 \le t \le 0 \\ 2+t, & -2 \le t \le -1 \\ 0, & \text{αλλού} \end{cases}$$

$$x_o = \frac{1}{2}\left[x(t) - x(-t)\right] = \begin{cases} \frac{1}{2}\left[t-0\right], & 0 \le t \le 1 \\ \frac{1}{2}\left[2-t-0\right], & 1 \le t \le 2 \\ \frac{1}{2}\left[0-(-t)\right], & -1 \le t \le 0 \\ \frac{-2-t}{2}, & -2 \le t \le -1 \end{cases}$$

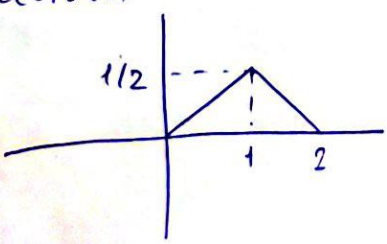
$$x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2-t, & 1 \le t \le 2 \\ 0, & \text{αλλού} \end{cases}$$

ΠΡΩΤΟΠΟΡΙΑ ΤΗΝ ΠΑΤΡΙΑ ΕΧΑΙΤΩ ΦΩΣ ΔΙΑΦΩΣΩ
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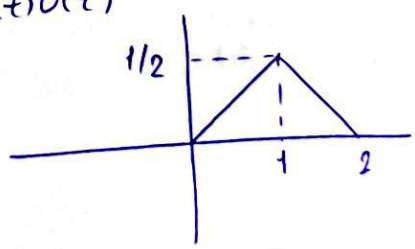




$x_e(t)u(t)$



$x_o(t)u(t)$



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1.5

2) $x(t) = e^{j(\pi t - 1)}$

$x(t+T) = x(t)$

$e^{j[\pi(t+T)-1]} = e^{j(\pi t - 1)}$

~~$e^{j\pi t}$~~ $e^{j\pi T}$ ~~e^{-j}~~ $= $ ~~$e^{j\pi t}$~~ ~~e^{-j}~~

$e^{j\pi T} = 1$

$\cos \pi T + j \sin \pi T = 1 + j \cdot 0$

~~π~~$T = 2k\pi$

$T = 2k$ $\Rightarrow$ για $k=1$ η περίοδος είναι $T=2$ και η συχνότητα $f_0 = \frac{1}{2}$ hz

3) $x(n) = \cos\left(\frac{8\pi}{7} n + 2\right)$

$x(n+N) = x(n)$

$\cos\left(\frac{8\pi}{7}(n+N) + 2\right) = \cos\left(\frac{8\pi n}{7} + 2\right)$

$\frac{4N}{7} = k$

$N = \frac{7}{4}k$ $\Rightarrow$ για $k=1$ η περίοδος είναι $N = \frac{7}{4}$ και η συχνότητα $f_0 = \frac{4}{7}$ hz

7) $x(n) = \cos\left(\frac{\pi}{4}\right) \cdot \cos\left(\frac{\pi n}{4}\right) = \frac{1}{2}\cos\left[\frac{n}{4} - \frac{\pi n}{4}\right] + \frac{1}{2}\cos\left[\frac{n}{4} + \frac{\pi n}{4}\right]$

$\frac{1}{2}\cos\left[(1-\pi)\frac{n+N}{n}\right] = \frac{1}{2}\cos\left[(1-\pi)\frac{n}{k}\right]$

και

$\frac{1}{2}\cos\left[(1+\pi)\frac{n+N}{n}\right] = \frac{1}{2}\cos\left[(1+\pi)\frac{n}{k}\right]$

$\frac{N}{n}(1-\pi) = 2k\pi$ και $\frac{N}{n}(1+\pi) = 2k\pi$, $k = 0, 1, 2$ } $\frac{k}{N} \neq 0$, $k = 1, 2$

$\frac{k}{N} = \frac{1-n}{8n}$ και $\frac{k}{N} = \frac{1+n}{8n}$, $k = 1, 2$ }

ΠΡΩΤΟΠΟΡΙΑ ΤΗΣ ΠΑΤΡΙΑΣ ΔΙΑΦΩΤΙΣΜΟΥ ΟΥΚ ΕΛΑΧΙΣΤΟΥ
ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΠΑΤΡΩΝ

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γ) $x(t) = Ae^{j(\omega_0 t+\theta)}$

$$tx = \lim_{t\to+\infty} \int_{-T}^{T} |Ae^{j(\omega_0 t+\theta)}|^2 = \int_{T}^{T} A^2 e^{2j(\omega_0 t+\theta)} dt$$

$$\lim_{T\to+\infty} A^2 \frac{e^{2j(\omega_0 t+\theta)}}{2j\omega_0} = \lim_{T\to+\infty} \frac{e^{\infty}}{2j\omega_0} - 0 = +\infty$$, Δ ενεργειακά σήματα

$$Px = \lim_{T\to+\infty} \frac{1}{2T} \int_{-T}^{T} |Ae^{j(\omega_0 t+\theta)}|^2 = \lim_{T\to+\infty} \int_{-T}^{T} A^2 e^{2j(\omega_0 t+\theta)} dt$$

$$= \lim_{T\to+\infty} \frac{1}{2T} A^2 \left[\frac{e^{2j(\omega_0 t+\theta)}}{2j\omega_0} \right]_{-T}^{T} = \lim_{T\to+\infty} \frac{e^{\infty}}{2\cdot\infty} = \infty$$, Δ σήματα ισχύος

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Ασκήσεις Κεφ. 2

Δωτήρης Γυγιώτης
22983

6/6

2.2) $y(t) = |x(t)| = |x(t-t_0)| = y_1(t)$

⇒ Είναι χρονικά αναλλοίωτο

$y(t) = |x(t)|$

σήματα εισόδου $x_1(t)$, $x_2(t)$

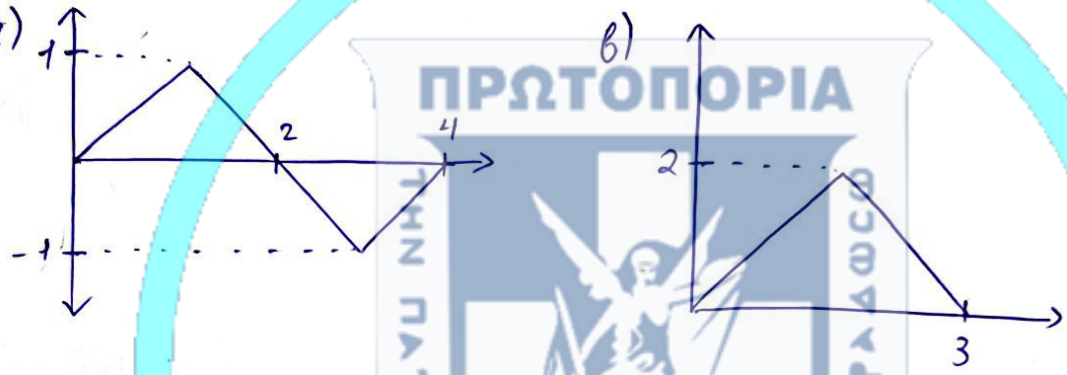
$S\{ax_1(t)+bx_2(t)\} = |ax_1(t)+bx_2(t)|$

$= ay_1(t)+by_2(t)$

⇒ Είναι γραμμικό σύστημα

Αιτιατό σήμα αφού η έξοδος εξαρτάται απ' την είσοδο.

2.4)



2.6) $y(t) = x(t) \cdot \cos(\omega_0 t)$

$S = \{ax_1(t)+bx_2(t)\}\cos(\omega_0 t)$

$= \{ax_1(t)\cos(\omega_0 t)+bx_2(t)\cos(\omega_0 t)\}$

$= ay_1(t)+by_2(t)$

⇒ είναι γραμμικό

Δεν είναι χρονικά αναλλοίωτο αφού υπάρχει το συνημίτονο.
Είναι αιτιατό αφού η έξοδος εξαρτάται απ' την είσοδο.

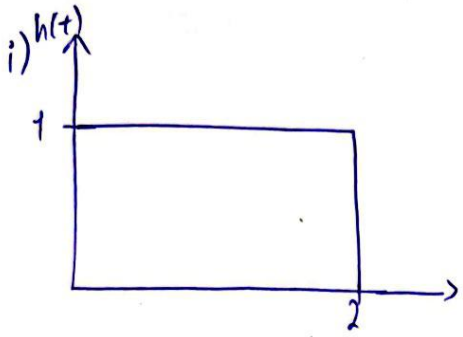
ΠΡΩΤΟΠΟΡΙΑ
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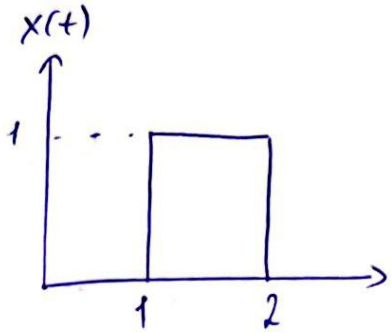
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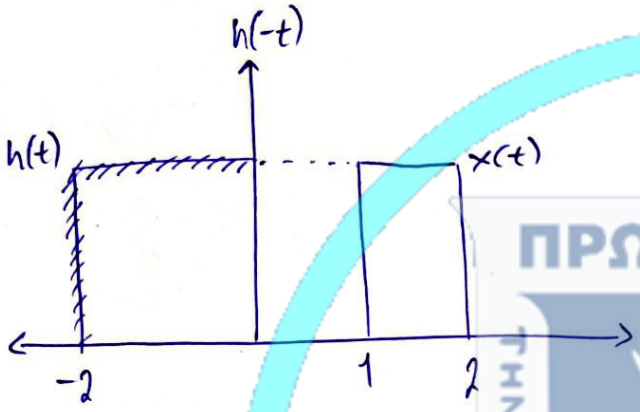
2.8) $h(t) = u(t) - u(t-2)$

$x(t) = u(t-1) - u(t-2)$

i)

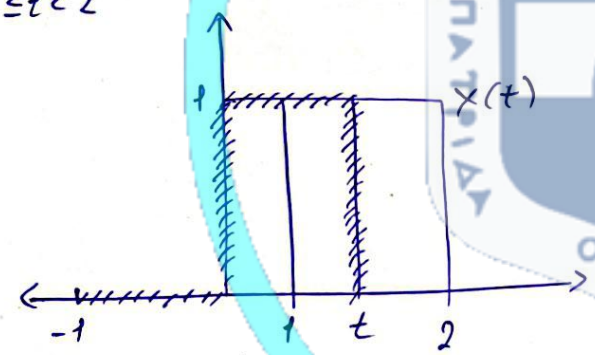






$\Rightarrow y(t) = 0$ για $x < 1$

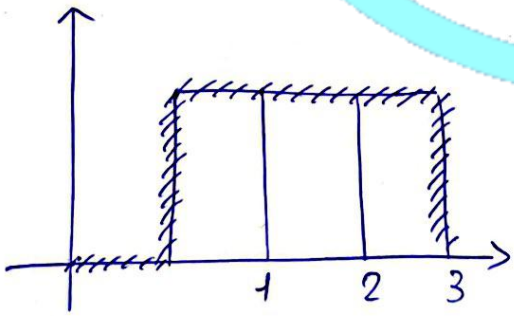
ii) $1 \le t < 2$



$y = (t-1)$

$\Rightarrow y = t-1$

iii) $2 \le t < 3$

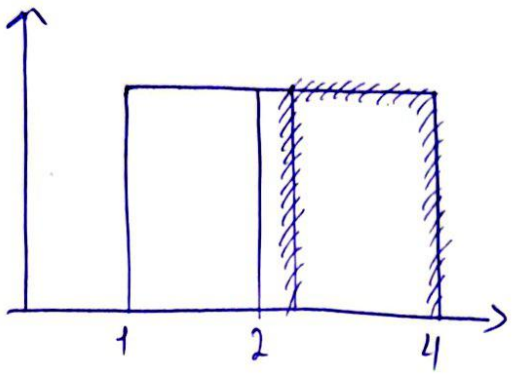


$y = 1$

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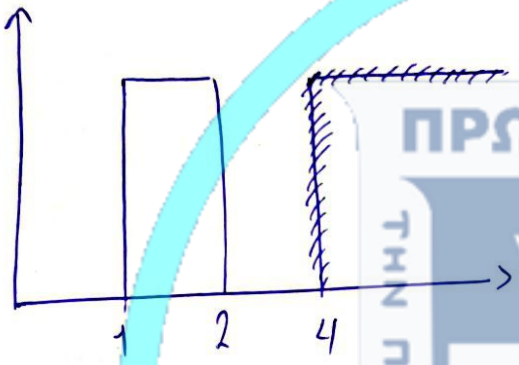
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$3 \le t < 4$



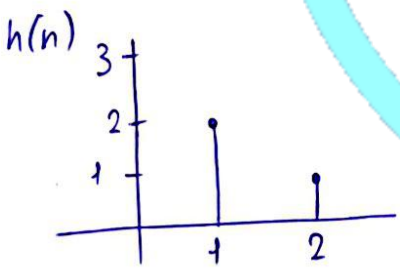
$y = |4 - t|$

γ) $4 \le t$

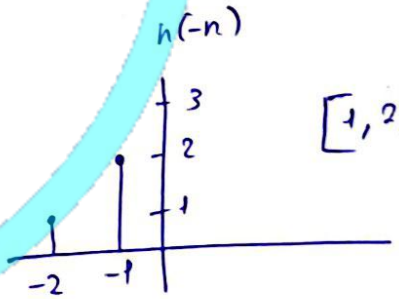


$y = 0$

2.12) $h(n) = [3, 2, 1]$
$\uparrow$
$x(n) = [1, 2, 3, 4]$
$\uparrow$



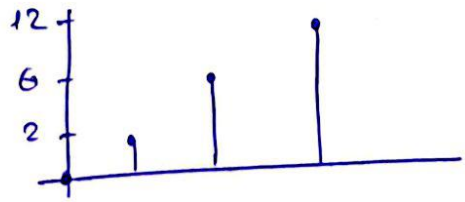




$[1, 2, 3]$
$\uparrow$

$h[3-u] = [0, 1, 2, 3]$

$x[u]\,h[3-u] = [1, 2, 3, 4] \cdot [0, 1, 2, 3] = [0, 2, 6, 12]$
$\uparrow$



ΠΡΩΤΟΠΟΡΙΑ ΤΗΝ ΠΑΤΡΙΔΑ ΠΑΡΑΔΩΣΩ ΕΛΛΑΤΤΩ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

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2.13) $\frac{dy(t)}{dt} + 10y(t) + 5 = x(t)$ (1)

Έχουμε σήματα εισόδου $x_1(t)$, $x_2(t)$

$$\frac{dy(t)}{dt} = x(t) - 10y(t) - 5$$

$$S\{ax_1(t) + bx_2(t)\} = c(ax_1(t) + bx_2(t))^2 \quad (a, b \in \mathbb{C})$$
$$= c(a^2x_1^2(t) + 2abx_1(t)x_2(t) + b^2y_2^2(t))$$
$$= a^2y_1(t) + 2cabx_1(t)y_2(t) + b^2y_2(t) + ay_1(t) + by_2(t)$$
$$= a, S\{x_1(t)\} + b, S\{y_2(t)\} \quad (2)$$

⇒ Το σύστημα (1) είναι μη-γραμμικό αφού δεν ισχύει ισότητα στο (2).

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Watermarkly

Ασκήσεις 1.1 (9/7)

Ιωάννης Γιαγκιώνης (77983)

(1) Περιοδικό σήμα $x(t)$, $T=8$

$C_1 = C_{-1} = 4$, $C_3 = C_{-3}^* = j2$

$C_n = 0 \ \forall n \neq \pm 1, \pm 3$

$\frac{2\pi}{T} = \frac{\pi}{2}$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n \cdot e^{jn\omega t} = 2je^{j(-3)\frac{\pi}{2}t} - 4e^{j(-1)\frac{\pi}{2}t} + 4e^{j\frac{3\pi}{2}t} + 4e^{j\frac{\pi}{2}t}$$

($\frac{2\pi}{T} = \frac{\pi}{2}$, $n=-3$, C_{-3}, C_{-1}, C_1, C_3)

$$= 2je^{-j\frac{3\pi}{2}t} - 4e^{-j\frac{\pi}{2}t} + 4e^{j\frac{3\pi}{2}t} + 4e^{j\frac{\pi}{2}t}$$

$A_n = 2|C_n| = 2|a_n - jb_n|$

$\Phi_n = \tan^{-1}\left(\frac{b_n}{a_n}\right)$

a) $2C_3 = 4j = a_3 - jb_3 \Rightarrow a_3 = 0, b_3 = -4$

$2C_1 = 4 = a_1 - jb_1 \Rightarrow a_1 = 4, b_1 = 0$

$\Rightarrow x(t) = 4\sin(2\pi t) - 4\cos(2\pi t)$

β) $x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cdot \cos(n\omega t) + b_n \cdot \sin(n\omega t)] = A_0 + \sum_{n=1}^{\infty} \cdot A_n \cdot \cos(n\omega t - \Phi_n)$

(3) α) $x_1(t) = 2 + 4\cos\left(\frac{\pi}{3}t\right) + \sin\left(\frac{2\pi}{3}t\right)$

$$= 2 + e^{-j\frac{\pi}{3}t} + e^{j\frac{\pi}{3}t} - \frac{1}{2j}e^{-j\left(\frac{2\pi}{3}t\right)} + \frac{1}{2j}e^{\left(\frac{2\pi}{3}t\right)}$$

$$= 2 + e^{j\frac{\pi}{3}t} + e^{j\frac{\pi}{3}t} + \frac{1}{2}\left(\frac{\sqrt{3}}{4} + j\frac{1}{4}\right)e^{-jt} + \frac{1}{2}\left(\frac{\sqrt{3}}{2} - j\frac{1}{2}\right)e^{jt}$$

$C_0 = 2$, $C_{-1} = C_1^*$, $C_{-4} = C_4^*$

δ) $x_4(t) = \sin^2(\pi t)$

$$= \frac{1}{2}[\sin(2\pi t) + 1]$$

$$= \frac{1}{4}e^{-j2\pi t} + \frac{1}{4}e^{j2\pi t} + \frac{1}{2}$$

$C_0 = \frac{1}{2}$, $C_{-1} = \frac{1}{4}j$, $C_1 = \frac{1}{4}$, $C_n = 0 \ \forall n \neq -1, 0, 1, \ldots$

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$\left(\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi\right)$

$$c_n = \frac{-j}{2\pi n}\left[e^{0} - e^{j n\pi}\right] + \frac{j}{2\pi n}\left[e^{-jn\pi} - e^{0}\right]$$

$$= \frac{-j}{2\pi n}\left[2 - (\cos(n\pi) + j\sin(n\pi)) - (\cos(n\pi) - j\sin(n\pi))\right]$$

$$= \frac{-j}{2\pi n}\left[2 - 2\cos(n\pi)\right]$$

$n=1$: $c_1 = \frac{-j}{2\pi\cdot 1}(2 - 2\cos\pi) = \frac{-2j}{\pi}$

$n=2$: $c_2 = \frac{-j}{2\pi\cdot 2}(2 - 2\cos(2\pi)) = 0$

$n=3$: $c_3 = \frac{-j}{2\pi\cdot 3}(2 - 2\cos(3\pi)) = \frac{-2j}{3\pi}$

$n=4$: $c_4 = \frac{-j}{2\pi\cdot 4}(2 - 2\cos(4\pi)) = 0$

$$\Rightarrow c_n = \begin{cases} \frac{-2j}{\pi n}, & n = 1,3,\ldots \text{ (περιττός)} \\ 0, & n = 2,4,\ldots \text{ (άρτιος)} \end{cases}$$

Σειρά Fourier:

$$x_2(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\omega t}$$

$$= \sum_{n=\pm 1,\pm 3,\ldots} \frac{-2j}{\pi n} \cdot e^{jn\pi t} = \sum_{n=-\infty}^{\infty} \frac{-2j}{\pi(2k+1)} \cdot e^{j(2k+1)\pi t}$$

$n = 2k+1$

$$= \left[\left(\frac{-2j}{\pi} e^{j\pi t} - \frac{2j}{3\pi} e^{j3\pi t} - \ldots\right) + \left(\frac{2j}{\pi} e^{-j\pi t} + \frac{2j}{3\pi} e^{-j3\pi t} + \ldots\right)\right]$$

(4) β) $x_2(t) = \begin{cases} -2, & 0 \le t < 1 \\ 2, & 1 \le t < 2 \end{cases}$

$$c_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x_2(t)\,dt = \frac{1}{4}\int_2^2 x_2(t)\,dt$$

$$= \frac{1}{4}\left\{\int_1^2 +2\,dt + \int_0^1 -2\,dt\right\} = \frac{1}{4}[t]_1^2 - \frac{1}{4}[t]_0$$

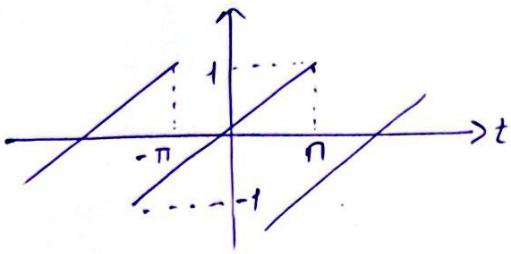
$$= \frac{1}{4}[2-1] - \frac{1}{4}[1-0] = 0$$

$$c_n = \frac{1}{4}\int_{-2}^{2} x_2(t)\cdot e^{-jn\omega t}\,dt = \frac{1}{4}\int_1^2 e^{-jn\omega t}\,dt - \frac{1}{4}\int_0^1 e^{-jn\omega t}\,dt$$

$$= \frac{1}{4}\left[\frac{e^{-jn\omega t}}{-jn\omega}\right]_1^2 - \frac{1}{4}\left[\frac{e^{-jn\omega t}}{-jn\omega}\right]_0^1$$

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) $x_1(t) = t^2, -\pi < t < \pi$



$$c_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x_1(t)\,dt = \frac{1}{2}\int_{-\pi}^{\pi} t^2\,dt = \frac{1}{2}\left[\frac{t^3}{3}\right]_{-\pi}^{\pi} = \frac{1}{2}\left(\frac{\pi}{3} - \frac{-\pi}{3}\right) = 0$$

$$c_n = \frac{1}{T}\int_{t_0}^{t_0+T} x_1(t)\cdot e^{-jn\omega t^2}\,dt = \frac{1}{2}\int_{-\pi}^{\pi} t\cdot e^{-jn\pi t^2}\,dt$$

$w = t^2 \Rightarrow w = 1$

$v = e^{-jn\pi t^2} \Rightarrow v = \dfrac{e^{-jn\pi t^2}}{-jn\pi}$

$$\Rightarrow c_n = \frac{1}{2}\left[t^2\frac{e^{-jn\pi t^2}}{-jn\pi}\right]_{-\pi}^{\pi} - \frac{1}{2}\int_{-\pi}^{\pi}\frac{e^{-jn\pi t^2}}{-jn\pi}\,dt$$

$$= \frac{1}{2}\left[t^2\frac{e^{-jn\pi t^2}}{-jn\pi} - \frac{e^{-jn\pi t^2}}{(-j)^2 n^2\pi^2}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\frac{-1}{jn\pi}\cdot e^{-jn\pi} + \frac{1}{n^2\pi^2}e^{-jn\pi} - \left(\frac{-1}{-jn\pi}\right)e^{jn\pi} - \frac{1}{n^2\pi^2}e^{jn\pi}\right]$$

$$c_n = \frac{-1}{n\pi}\cdot\frac{1}{2}\left(e^{jn\pi} + e^{-jn\pi}\right) - \frac{1}{n^2\pi^2}\cdot\frac{j}{2}\left(e^{jn\pi} - e^{-jn\pi}\right)$$

$$= \frac{-1}{n\pi}\cos(1\cdot\pi) - \frac{1}{n^2\pi^2}j\sin(n\cdot\pi)$$

$n=1$: $c_1 = \dfrac{-1}{1\cdot\pi}\cos(\pi) - \dfrac{1}{1^2\pi^2}j\sin(\pi) = \dfrac{1}{\pi}$

$n=2$: $c_2 = \dfrac{-1}{2\pi}\cos(2\pi) - \dfrac{1}{4\pi^2}j\sin(2\pi) = \dfrac{-1}{2\pi}$

$n=3$: $c_3 = \dfrac{-1}{3\pi}\cos(3\pi) - \dfrac{1}{9\pi^2}j\sin(3\pi) = \dfrac{1}{3\pi}$

$n=4$: $c_4 = \dfrac{-1}{4\pi}\cos(4\pi) - \dfrac{1}{16\pi^2}j\sin(4\pi) = \dfrac{-1}{4\pi}$

$$c_n = \begin{cases} \dfrac{1}{n\pi}, & n = 1,3,5,\ldots \text{ (περιττός)} \\ \dfrac{-1}{n\pi}, & n = 2,4,6,\ldots \text{ (άρτιος)} \end{cases}$$

$$\Rightarrow x_1(t) = \sum_{n=-\infty}^{\infty} c_n\cdot e^{jn\omega t}$$

$$= \frac{1}{\pi}\sum_{n=\pm1,\pm3}\frac{1}{n}\cdot e^{jn\pi t} - \frac{1}{\pi}\sum_{n=\pm2,\pm4}^{\infty} e^{jn\pi t}$$

$$= \frac{1}{\pi}\sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty}\frac{(-1)^{n+1}}{n}\cdot e^{jn\pi t}$$

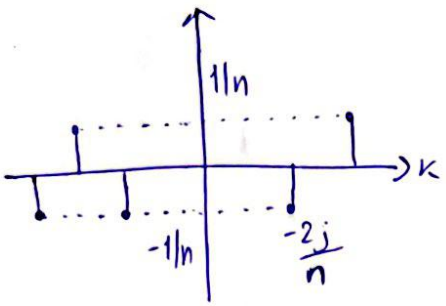
ΠΡΩΤΟΠΟΡΙΑ ΠΑΡΑΔΟΣΩ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

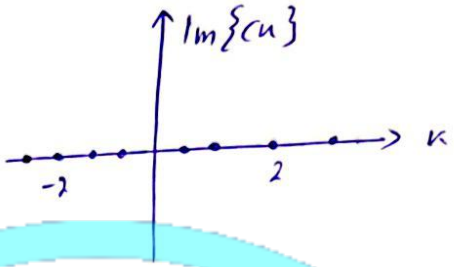
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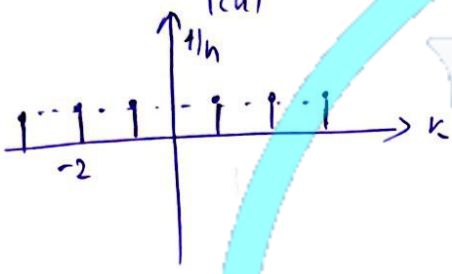
(7) 4Β) $c_n = \left(\frac{-2}{\pi}\right)^n \cdot 2^{n+1}$

$Re\{c_n\} = \frac{(-2)^n}{2^{n+1}}$, $Im\{c_n\} = 0$ $\forall n$, $|c_n| = \sqrt{Re^2 + Im^2} = \frac{1}{2^{n+1}} \cdot \frac{1}{\pi}$

$Arg\{c_n\} = \tan^{-1}\left(\frac{Im}{Re}\right) = 0 \; \forall n$







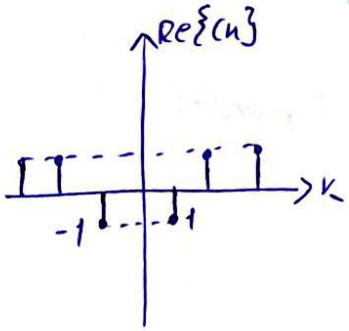


6Β) $x_1(t) = t^2$, $-\pi < t < \pi$

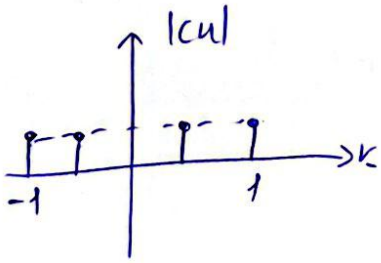
$c_n = \frac{(-1)^n}{\pi} \cdot \frac{1}{n+1} + 0j$

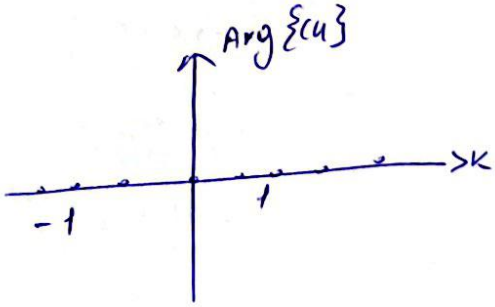
$\Rightarrow Re\{c_n\} = \frac{(-1)^n}{n+1}$, $Im\{c_n\} = 0$ $|c_n| = \sqrt{Re^2 + Im^2} = \frac{1}{n+1} \cdot \frac{1}{\pi}$

$Arg\{c_n\} = \tan^{-1}\left(\frac{Im}{Re}\right) = 0 \; \forall n$









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Watermarkly

ΕΙΣ 1.2 β/α

α) $x(t) = t^3$

$x(-t) = x(t)$ (Η) $x(-t) = -x(t)$

ΑΡΤΙΟ ΠΕΡΙΤΤΟ

$x(-t) = (-t)^3 = -t^3 = -x(t) \Rightarrow$ περιττό

β) $x(t) = t + t^2$

$x(-t) = (-t) + (-t)^2 = t^2 - t \Rightarrow$ τίποτα απ'τα δύο

γ) $x(t) = t\cos t$

$x(-t) = (-t)\cos(-t) = -t\cos t = -x(t) \Rightarrow$ περιττό

δ) $x(t) = t^2\cos t$

$x(-t) = (-t)^2\cos(-t) = t^2\cos t \Rightarrow$ άρτιο

ε) $x(t) = t^2\sin t$

$x(-t) = (-t^2)\sin(-t) = -t^2\sin t \Rightarrow$ περιττό

στ) $x(t) = |2t|$

$x(-t) = |-2t| = |2t| \Rightarrow$ άρτιο

ζ) $x(t) = t|t|$

$x(-t) = -t|-t| = -t|t| = x(t) \Rightarrow$ περιττό

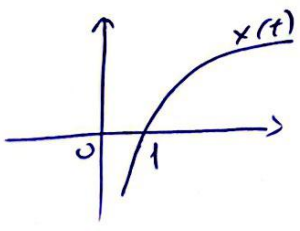
η) $x(t) = e^t$

$x(-t) = e^{-t} \Rightarrow$ τίποτα απ'τα δύο

θ) $x(t) = e^{t^2}$

$x(-t) = e^{(-t)^2} = e^{t^2} = x(t) \Rightarrow$ άρτιο

ι) $x(t) = \ln t$ (μόνο για $t > 0$) $\Rightarrow$ τίποτα απ'τα δύο



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Watermarkly

(3) Β) $x_2(t) = 2t+1,\ 0 \le t \le \pi$

$$c_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x_1(t)\,dt = \frac{1}{\pi}\int_0^{\pi} t\,dt = \frac{1}{\pi}\left[\frac{t^2}{2}\right]_0^{\pi} = \frac{1}{2}\left(\frac{\pi^2}{2}-\frac{0}{2}\right) = \frac{\pi^2}{4}$$

$$c_n = \frac{1}{T}\int_{t_0}^{t_0+T} x_1(t)\cdot e^{-jn\omega t}\,dt = \frac{1}{\pi}\int_0^{\pi} e^{-jn2t}\,dt$$

$\omega = t \Rightarrow \omega = 1$

$V = e^{-jn2t} \Rightarrow V = \dfrac{e^{-jn2t}}{-jn2t}$

$$\Rightarrow c_n = \frac{1}{\pi}\left[t\cdot\frac{e^{-jn2t}}{-jn2t}\right]_0^{\pi} - \frac{1}{\pi}\int_0^{\pi}\frac{e^{-jn2t}}{-jn2}\,dt$$

$$= \frac{1}{\pi}\left[t\cdot\frac{e^{-jn2t}}{-jn2} - \frac{e^{-jn2t}}{(-j)^2 n^2 2^2}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{-1}{jn2}\cdot e^{-jn2} + \frac{1}{n^2\pi^2}e^{-jn\pi} - \left(\frac{-1}{-jn\pi}\right)e^{jn2}\cdot\frac{-1}{n^2 2^2}e^{jn2}\right]$$

$$c_n = \frac{-1}{n2}\cdot\frac{1}{\pi}\left(e^{jn2}+e^{-jn2}\right) - \frac{1}{n^2 2^2}\cdot\frac{1}{\pi}\left(e^{jn2}-e^{-jn2}\right)$$

$$= \frac{1}{n2}\cos(n2) - \frac{1}{n^2 2^2}\,j\sin(n2)$$

$n=1$: $c_1 = \frac{-1}{1\pi}\cos(\pi) - \frac{1}{1^2\pi^2}\,j\sin(\pi) = \frac{1}{\pi}$

$n=2$: $c_2 = \frac{-1}{2\pi}\cos(2\pi) - \frac{1}{4\pi^2}\,j\sin(2\pi) = \frac{-1}{2\pi}$

$n=3$: $c_3 = \frac{-1}{3\pi}\cos(3\pi) - \frac{1}{9\pi^2}\sin(3\pi) = \frac{1}{3\pi}$

$$c_n = \begin{cases} \frac{1}{n\pi}, & n = 1,3,5,\ldots \text{ (περιττοί)} \\ \frac{-1}{n\pi}, & n = 2,4,6,\ldots \text{ (άρτιοι)} \end{cases}$$

$$\Rightarrow x_2(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega t}$$

$$= \frac{1}{\pi}\sum_{n=\pm1,\pm3}\frac{1}{n}e^{jn2t} - \frac{1}{\pi}\sum_{n=\pm2,\pm4}\frac{1}{n}e^{jn2t}$$

$$= \frac{1}{\pi}\sum_{\substack{n\neq 0 \\ n=-\infty}}^{\infty}\frac{(-1)^{n+1}}{n}\cdot e^{jn2t}$$

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Watermarkly

ΕΙΣ 1.3 (8/8)

δ) $x_3(t) = x(2t-2) + e^{j2\omega t} - x(2-t)$

$x(2t-2) \xrightarrow{FS2} c_n \cdot e^{-jn\omega a} = c_n \cdot e^{-jn\omega 2}$ (f_n)

$\Rightarrow e^{j2\omega t} \cdot x(t) \xrightarrow{FS3} c_{n-a} = c_{n-2}$ (f_n)

$\Rightarrow x(-(t-2)) \xrightarrow{FS5} f_{-n} = c_{-n} \cdot e^{jn\omega 2}$ (g_n)

$\Rightarrow x(2t-2) + e^{j2\omega t} \cdot x(2-t) \xrightarrow{FS1} f_n + f_n \cdot g_n$

$\Rightarrow \underbrace{c_n \cdot e^{-jn\omega 2} + c_{n-2} \cdot c_n e^{jn\omega 2}}_{c_{3\cdot n}}$

ε) $x_5(t) = \underbrace{e^{j2\omega t}}_{FS3} \int_{<T>} x(-u)\, y(t-u)\, du = \underbrace{x(t)}_{FS5} * \underbrace{y(t)}_{FS7}$

$e^{j2\omega t} x(t) \xrightarrow{FS3} c_{n-a} = c_{n-2}$

$x(t) \xrightarrow{FS5} c_{-n}$ (f_n)

$x(-t) * y(t) \xrightarrow{FS7} T f_n d_n = T c_n d_n \; (= c_n)$

$x_5(f) = f_n \cdot c_n = c_{n-2} \cdot c_n$

$x(t) \xrightarrow{FS} c_n, \; y(t) \xrightarrow{FS} d_n$

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ε) $x_7(t) = \underbrace{x(2t-1)}_{FS2} \cdot \underbrace{e^{j\omega t}}_{FS3} \cdot \underbrace{x(-t)}_{FS5}$

$\Rightarrow x(2t-1) \xrightarrow{FS2} c_n \cdot e^{-jn\omega a} = c_n \cdot e^{-jn\omega 1}$ (f_n)

$\Rightarrow e^{j\omega t}\; x(t) \xrightarrow{FS1} c_{n-a} = c_{n-1}$ (f_n)

$\Rightarrow x(-t) \xrightarrow{FS5} f_n = c_{-n} \cdot e^{jn\omega}$ (g_n)

$\Rightarrow x_7 \; c_n \cdot e^{-jn\omega \cdot 1} \cdot c_{n-1} \cdot c_n e^{jn\omega}$

η) $x_8(t) = \underbrace{x(2t-1)}_{FS2} \cdot y(-2t)$, $x(t) \xrightarrow{FS} c_n$, $y(t) \xrightarrow{FS} l_n$

$x(2t-1) \xrightarrow{FS2} c_n \cdot e^{-jn\omega a} = c_n \cdot e^{-jn\omega \cdot 1}$ (f_n)

$\Rightarrow x_8(t) = x(2t-1)\, y(-2t) = f_n \cdot d_n = c_n \cdot e^{-jn\omega \cdot 1} \cdot d_n$

Watermarkly

1) $x_{10}(t) = \underbrace{e^{-jwt}}_{FS3} \underbrace{\frac{d}{dt}}_{FS4} \left[\underbrace{x(1-t)}_{FS2} \right]_{FS5}$ $\quad x(t) \xrightarrow{FS} c_n$

$\Rightarrow e^{-jwt} x(t) \xrightarrow{FS3} f_{n+1} = c_{n+1} \cdot e^{-j(n+1)w}$ (f_n)

$\Rightarrow x(-(t-1)) \xrightarrow{FS5} f_{-n} = c_{-n} \cdot e^{jnw1}$ (g_n)

$x_{10}(t) = e^{-jwt} \frac{d}{dt} [x(1-t)] \xrightarrow{FS4} \underbrace{jnw \cdot g_n = jnw c_{n+1} \cdot e^{-j(n+1)w}}_{c_{gn}}$

2) $x(t)$ & $y(t) = x'(t) = \frac{dx(t)}{dt}$

$\int_{<T>} x(t)\,dt = 3$

$x(t) \xrightarrow{FS} c_n, y(t) \xrightarrow{FS} d_n$

$x(t) = c_n = 3$

4) i) $x(t) \in \mathbb{R}$ και άρτιο

ii) $x(t)$ περιοδικό $T=3$

iii) $c_n = 0$ για $|n| > 1$

iv) $\int_0^3 |x(t)|^2 dt = 3$

Από (iii) $c_0, c_{-1}, c_1 \neq 0 \Rightarrow x(t) = c_0 + c_{-1} \cdot e^{-jwt} + c_1 \cdot e^{jwt}$ $(w = \frac{2\pi}{T} = \pi)$

Από (i) $\Rightarrow c^* = c_1$

Από (iv) ισχύει για $x(t-1) \Rightarrow \int_0^3 |x(t)|^2 dt = 3 \Leftrightarrow \int_0^3 |y(t)|^2 dt = 3$

$\Rightarrow 3 = \int_0^3 |y(t)|^2 dt = 3 \sum_{n=-\infty}^{\infty} |d_n|^2 = 3[|d_{-1}|^2 + |d_1|^2] = 9|d_1|^3$

$d_1 = \pm j \frac{1}{\sqrt{3}}$ $(\in \mathbb{I})$

$c_0 = 0, c_1 = d_1 e^{j\pi} = \pm j \frac{1}{\sqrt{3}} (-1) = \mp j \frac{1}{\sqrt{3}} = c^*$

$x(t) = c_1 \cdot e^{-j\pi t} + c_1 \cdot e^{j\pi t} = \pm \left(j \frac{1}{\sqrt{3}} e^{-j\pi t} - j \frac{1}{\sqrt{3}} \cdot e^{j\pi t} \right) = \pm \sqrt{3} \cdot \sin \pi t$

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a) $1 + \sum_{n=1} \frac{1}{(2n+1)^4} = 1 + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \dots = \frac{\pi^4}{96}$

$$x_1(t) = \begin{cases} t, & -\frac{\pi}{2} < t < \frac{\pi}{2} \\ \pi - t, & \frac{\pi}{2} < t < \frac{3\pi}{2} \end{cases}$$

$$P_x = \frac{4}{\pi^2} \sum_{n=-\infty}^{\infty} \frac{1}{(2|n|+1)^4} \cdot = \frac{16}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}$$

$$\frac{1}{2\pi} \int_{\pi-t}^{t} |x(t)|^4 dt = \frac{16}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}$$

$$\int_{\pi-t}^{t} |x(t)|^4 dt = \int_{\pi-t}^{0} 1\, dx + \int_{0}^{t} 1\, dx + 0 = 2\pi$$

$$\frac{\pi^2}{16} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} = 1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

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Ασκήσεις 1.4 (5/5)

① $x(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\pi t}$

$c_0 = \frac{1}{2}, c_1 = j, c_{-1} = -j, c_2 = -\frac{1}{2}, c_{-2} = -\frac{1}{2}$

$h(t) = e^{-t} u(t) \quad (u(t)), \quad y(t) = ?$

$y(t) = \sum_{n=-\infty}^{\infty} d_n e^{jn\pi t} = \sum_{n=-\infty}^{\infty} H(jn\pi) c_n \cdot e^{jn\pi t}$

$H(jn\pi) = \int_{-\infty}^{\infty} h(t) \cdot e^{-jnwt} dt = \int_{0}^{\infty} e^{-t} \cdot e^{-jn\pi t} dt = \frac{1}{1 - jn\pi}$

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$d_0 = H(j0\pi)c_0 = \frac{1}{2}\cdot\left(\frac{1}{2}\right) = \frac{1}{4}$

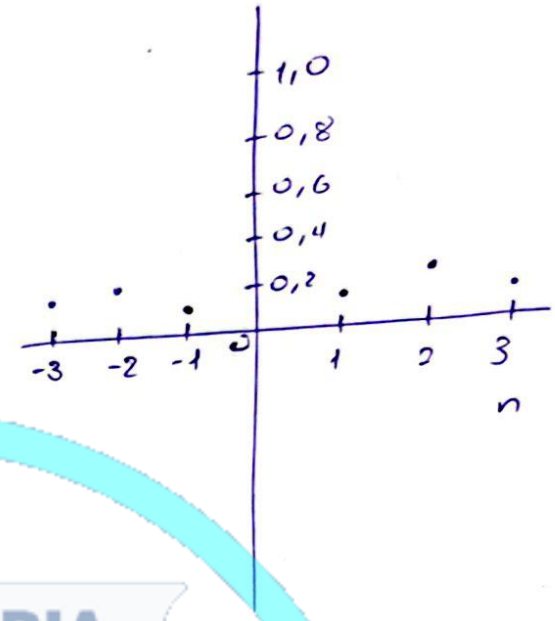
$d_1 = H(j\pi)c_1 = \frac{1}{1+j\pi}\left(j\right) = \frac{1(1-j\pi)}{1+\pi^2}$

$d_{-1} = H(-j\pi)c_{-1} = \frac{1}{1-j\pi}\left(j\right) = \frac{j(1+j\pi)}{1+\pi^2}$

$d_2 = H(j2\pi)c_2 = \frac{1}{1+2j\pi}\left(j\frac{1}{2}\right) = \frac{1\cdot(1-2j\pi)}{2+8\pi^2}$

$d_{-2} = H(-j2\pi)c_{-2} = \frac{1}{1-2j\pi}\left(-j\frac{1}{2}\right) = \frac{-j(1+2j\pi)}{2+8\pi^2}$

$d_3 = H(j3\pi)c_3 = \frac{1}{1+3j\pi}\left(j\frac{1}{3}\right) = \frac{j(1+3j\pi)}{3+27\pi^2}$



$d_n = 0 \;\forall n \neq 0, \pm1, \pm2, \pm3$

(2) α) $h(t) = e^{-2|t|}$, $y(t) = ?$

$$x(t) = \begin{cases} 1, & -1/2 \le t < 1/2 \\ 0, & 1/2 \le t < 3/2 \end{cases}$$

$$h(t) = e^{-2|t|} \equiv \begin{cases} e^{-2t}, & t \ge 0 \\ e^{2t}, & t < 0 \end{cases}$$

$$y(t) = \sum_{n=-\infty}^{\infty} H(jn\omega)c_n \cdot e^{jn\omega t}$$

$$H(jn\omega) = \int_{-\infty}^{\infty} h(t)\cdot e^{-jn\omega t}\,dt$$

$$\Rightarrow \int_0^{\infty} e^{-2t}\cdot e^{-jn\omega t}\,dt = \frac{1}{2+jn\omega}$$

$$\Rightarrow \int_{-\infty}^{0} e^{2t}\cdot e^{-jn\omega t}\,dt = \frac{1}{2-jn\omega}$$

$$c_0 = \int x(t)\,dt = \int_{-1/2}^{1/2} dt = \int_{3/2}^{1/2} dt$$

$$c_n = \int_{-1/2}^{1/2} e^{2t}\cdot e^{-jn\omega t}\,dt + \int_{3/2}^{1/2} e^{-2t}\cdot e^{-jn\omega t}\,dt = \left[\frac{1}{2-jn\omega t}\right]_{-1/2}^{1/2} + \left[\frac{1}{2+jn\omega}\right]_{3/2}^{1/2}$$

$$\omega = \frac{2\pi}{T} = \frac{4\pi}{3}$$

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$H(j\omega) = \sin(3\omega)$

$$x(t) = \begin{cases} 1, & 0 \le t < 2 \\ -1, & 2 \le t < 4 \end{cases}$$

$$y(t) = \sum_{n=-\infty}^{\infty} H(jn\omega) \cdot C_n \cdot e^{jn\omega t}$$

$$C_0 = \frac{1}{T}\int_{<T>} x(t)dt = \frac{1}{4}\int_0^2 -dt + \frac{1}{4}\int_2^4 dt$$

$$= \frac{1}{4}[-t]_0^2 + \frac{1}{4}[t]_2^4 = \frac{1}{4}[-2+0+4-2] = 0$$

$$C_n = \frac{1}{T}\int_{<T>} x(t) \cdot e^{-jn\omega t} dt$$

$$= \frac{1}{4}\int_0^2 -e^{-jn\omega t} + \frac{1}{4}\int_2^4 e^{-jn\omega t} dt \quad \left(\omega = \frac{2\pi}{T} = \frac{\pi}{2}\right)$$

$$= \frac{1}{4jn\omega}\left\{\left[e^{-jn\pi t/4}\right]_0^2 - \left[e^{-jn\pi t/4}\right]_4^8\right\}$$

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Ασκήσεις 2.1 (8/8)

Ιωάννης Γιγιώτης
(32983)

(1) $x(t) \Rightarrow u(t)$ $\quad x(t) = 0$ για $t < 0$

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

$$X(\omega) \int_0^{\infty} e^{-j\omega t} dt \Rightarrow X(\omega) = \frac{-1}{j\omega} \left[e^{-j\omega t} \right]_0^{\infty}$$

$$= \frac{-1}{j\omega} \left[\lim_{t \to \infty} e^{-j\omega t} - e^{0} \right]$$

$$\lim_{t \to \infty} e^{-j\omega t} = \lim_{t \to \infty} \left[\cos(\omega t) - j \sin(\omega t) \right] = 0$$

Άρα ∃ μετασχηματισμός fourier

$$X(\omega) = \frac{-1}{j\omega} + \pi \delta(\omega)$$

(3) a) $x_1(t) = \cos t \, u(t)$

$$X_1(\omega) = F\{x_1(t)\} = \int_{-\infty}^{+\infty} \cos t \, u(t) \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} \frac{e^{jt} - e^{-jt}}{j} \cdot e^{-j\omega t} dt = \int_0^{\infty} \frac{1}{j} \left(e^{j(1-\omega)t} - e^{-j(2+\omega)t} \right) dt$$

$$= \frac{1}{j} \left[\frac{e^{j(1-\omega)t}}{j(1-\omega)} - \frac{e^{-j(1+\omega)}}{-j(1+\omega)} \right]_0^{+\infty}$$

$$= \lim_{L \to \infty} \frac{1}{j} \left[\frac{e^{j(1-\omega)L}}{j(1-\omega)} - \frac{e^{j(1-\omega)}}{j(1-\omega)} + \frac{e^{-j(1+\omega)}}{j(1+\omega)} - \frac{e^{-j(1+\omega)}}{j(1+\omega)} \right]$$

$$= \left(\frac{e^{j(1-\omega)}}{1-\omega} + \frac{e^{-j(1+\omega)}}{1+\omega} \right), \quad \omega > 1$$

(6)

$$x(\omega) = \begin{cases} -1, & -1 < \omega < 0 \\ +1, & 0 < \omega < 1 \\ 0, & \text{αλλού} \end{cases}$$

$$x(\omega) = -[u(\omega+0) - u(\omega-1)]$$

$$x(\omega) = [u(\omega+1) - u(\omega+0)]$$

$\Rightarrow x(t) = \frac{\sin t}{t}$

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(8) β) $x_2(\omega) = \delta(\omega-\pi) + 2\delta(\omega+\pi) + \pi\delta(\omega-\pi)$

$$x_2(t) = \frac{1}{2\pi} e^{j\pi t} + \frac{1}{\pi} e^{-j\pi t} + \frac{1}{2} e^{j\pi t}$$

(9) $x(t) = e^{-3t} u(t)$

α) $S_x(\omega) = |x(\omega)|^2 = x(\omega) \cdot x(\omega) = \frac{1}{j+\omega^2}$

β) $R_x(t) = F^{-1}\{S_x(t)\} = F^{-1}\left\{\frac{1}{\omega^2+9}\right\} = F^{-1}\left\{\frac{1/3}{\omega+3} - \frac{1/3}{\omega-3}\right\} = \frac{1}{3} e^{-3|\tau|}$

γ) $E_x = R_x(0) = \frac{1}{3}$

(11) $x(t) = \cos(2t)$

$$P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T\to\infty} \int_0^{T/2} dt = \lim_{T\to\infty} \frac{T}{2} = \infty \rightarrow \text{σήμα ισχύος}$$

α) $P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \cos^2(2t)\, dt = \lim_{T\to\infty} \frac{1}{2T} \int_0^{T/2} \frac{1+\cos(4t)}{2} dt$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_0^{T/2} \left(\frac{1}{2} + \frac{\cos(4t)}{2}\right) dt = \lim_{T\to\infty} \frac{1}{2T} \left[\frac{t}{2} + \frac{\sin(4t)}{8}\right]_0^{T/2} = \frac{1}{8}$$

β) $\bar{R}_x(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x^*(t-\tau) dt = \lim_{T\to\infty} \frac{1}{T} \int_0^{T/2} \cos(2t) \cdot \cos(2t-\tau) dt$

$$= \lim_{T\to\infty} \frac{1}{T} \int_\tau^{T/2} dt = 1$$

γ) $\bar{S}_x(\omega) = F\{\bar{R}_x(t)\} = \pi.\delta.\omega$

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(13) β) $x_2(t) = e^{-3t}$

$$y_2(\omega) = F\{e^{-3t}u(t)\} = \frac{1}{3+j\omega}$$

$$x(\omega) = F\{e^{-3t}u(t)\} = y_2(\omega) = \frac{1}{3+j\omega} \quad (\in \mathbb{R})$$

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Ασκήσεις 2.2 (10/10)

Σωτήρης Γιαγκιώτης (22983)

(1)

a) $x_1(t) = e^{-3|t|} = e^{-3t}u(t) + e^{3t}u(t)$

$y_1(\omega) = F\{e^{-3t}u(t)\} = \frac{1}{3+j\omega}$

$\overset{FTS}{\Rightarrow} y_2(\omega) = F\{e^{3t}u(-t)\} = y_1(-\omega) = \frac{1}{3-j\omega}$

$\overset{FT1}{\Rightarrow} X(\omega) = F\{e^{-3t}u(t) + e^{-3t}u(t)\}$

$= y_1(\omega) + y_2(\omega) = \frac{1}{3+j\omega} + \frac{1}{3-j\omega} = \frac{6}{6+\omega^2} \quad (\in \mathbb{R})$

β) $x_2(t) = 2\sin(2t)\cdot\cos(2t)$

$\overset{FT8}{\Rightarrow} X(\omega) = F\{\sin(2t)\cos(2t)\} = \frac{1}{2\pi} F\{\sin(2t)\} + F\{\cos(2t)\}$

$\left(= \frac{1}{2\pi}\int_{-\infty}^{\infty} x_1(\theta)x_2(\omega-\theta)d\theta\right)$

$= \frac{1}{2\pi}\left\{\frac{\pi}{j}\left[\delta(\omega-2) - \delta(\omega+2)\right]\cdot\pi\left[\delta(\omega-1) + \delta(\omega+1)\right]\right\}$

γ) $x_4(t) = \sin(-2t)\cdot u\cdot t$

$= \frac{1}{2}(e^{-2jt} + e^{2jt})u\cdot t$

$\overset{FT1, FT3}{\Rightarrow} X(\omega) = F\{\frac{1}{2}\cdot e^{-2jt}u(t)\} + F\{\frac{1}{2}e^{2jt}u(t)\}$

$= \frac{1}{2}y(\omega-2) + \frac{1}{2}y(\omega+2) \quad (y(\omega) = F\{u(t)\})$

$= \frac{1}{2}\cdot\frac{1}{j(\omega-2)} + \frac{1}{2}\pi\delta(\omega-2) + \frac{1}{2}\cdot\frac{1}{j(\omega+2)} + \frac{1}{2}\pi\delta(\omega+2)$

(2)

$sgn(t) = \begin{cases} -1, & t<0 \\ +1, & t>0 \end{cases}$

$sgn = -u(t) + u(t) = u(t) - u(-t)$

$y(\omega) = F\{u(t) - u(-t)\} = \frac{e^{-j\omega} - e^{j\omega}}{j\omega}$

$\overset{FT9}{\Rightarrow} X(\omega) = \frac{1}{j\omega}y(\omega) = \frac{e^{-j\omega} - e^{j\omega}}{\omega^2}$

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(4)

β) $x_2(t) = \underbrace{e^{j2t}}_{FT3} \cdot \underbrace{x(t)}_{FT2} + \underbrace{y(t-1)}_{FT4}$

$x(t) \xrightarrow{FT} X(\omega)$, $y(t) \xrightarrow{FT} Y(\omega)$

$\Rightarrow e^{j2t} \cdot x(t) \xrightarrow{FT3} Z_3(\omega) = e^{j2\omega} \cdot X(\omega)$ (1)

$\Rightarrow y(t-1) \xrightarrow{FT2} e^{-j\omega} X(\omega) \leftarrow Z_1(\omega)$ (2)

$\Rightarrow X_2(\omega) = (1) + (2)$

δ) $x_4(t) = \int_{-\infty}^{\infty} x(-u)\, y(t-u)\, du = \underbrace{x(-t)}_{FT5} \cdot \underbrace{y(t)}_{FT7}$

$x(t) \xrightarrow{FT} X(\omega)$, $y(t) \xrightarrow{FT} Y(\omega)$

$\Rightarrow x(-t) \xrightarrow{FT5} X(-\omega)$

$\Rightarrow x(-t) \cdot y(t) \xrightarrow{FT7} X(-\omega) \cdot Y(\omega) \leftarrow Z(\omega)$

$\Rightarrow X_5(\omega) = X(-\omega) \cdot Y(\omega)$

ε) $x_7(t) = \underbrace{x(2t-1)}_{FT2} \cdot \underbrace{e^{jt}}_{FT3} \underbrace{x(-t)}_{FT5}$

$x(t) \xrightarrow{FT} X(\omega)$

$\Rightarrow x(2t-1) = x\left(2\left(t-\tfrac{1}{2}\right)\right) \xrightarrow{FT6} \tfrac{1}{2} Z\left(\tfrac{\omega}{2}\right) = \tfrac{1}{2} e^{-j\omega} X(\omega)$ (1)

$\Rightarrow e^{jt} x(-t) \xrightarrow{FT3} Z(\omega)$ (2)

$x_7(t) = (1) \neq (2)$

$x(1-t) = x(-(t-1)) \xrightarrow{FT5} Z(-\omega) = e^{j\omega} X(-\omega)$ (2)

ι) $x_{10}(t) = \underbrace{e^{-j2t}}_{FT3} \underbrace{\frac{d}{dt}}_{FT4} \left[\underbrace{x(1-t)}_{FT2} \underbrace{}_{FT5} \right]$

$\Rightarrow e^{-j2t} \xrightarrow{FT3} Z_1(\omega-2) = e^{-j2(\omega-2)} X(\omega-2)$

$\Rightarrow \frac{d}{dt}[x(1-t)] \xrightarrow{FT4} j\omega X(\omega) + e^{j\omega} X(\omega)$

$\Rightarrow X_{10}(\omega) = e^{-j2(\omega-2)} X(\omega-2)\, j\omega X(\omega) + e^{j\omega} X(\omega)$
(2)

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(5) β) $x_2(t) = \frac{4e^{3jt}}{(t-2)^2+9}$ → 21.1/6

→ FT 2/2.2.1

$\Rightarrow Y_2(\omega) = \frac{8\pi\delta(\omega-3)}{e^{j2\omega} \times \omega+9}$

στ) $x_6(t) = \left[e^{-jt} \cdot \frac{\sin 2t}{t} \right]$ ← FT9

(FT3)

$x(t) = \frac{\sin 2t}{t} \Rightarrow X(\omega) = \pi \underbrace{[u(\omega+2) - u(\omega-2)]}_{Z(\omega)}$

$\Rightarrow e^{-jt} \cdot \frac{\sin 2t}{t} \xrightarrow{FT3} Z(\omega+1)$

$Y_6 = \pi [u(\omega+1) - u(\omega-3)] \xleftarrow{FT9}$

(6) γ) $X_3(\omega) = \frac{2e^{-3j\omega}}{1+j(2-\omega)}$ → FT2

(FT3 ↗, ↑ FT5)

$\frac{2}{1+j\omega} \xrightarrow{FT1} 2e^{-t} \cdot u(t) \leftarrow Z_1(t)$

$\Rightarrow \frac{2e^{-j\omega}}{1+j\omega} \xrightarrow{FT2} Z_1(t-1) = 2e^{-(t-1)} \cdot u(t-1) \leftarrow Z_2(t)$

$\Rightarrow \frac{2e^{j\omega}}{1-j\omega} \xrightarrow{FT5} Z_2(-t) = 2e^{t+1} \cdot u(-t-1) \leftarrow Z_3(t)$

$\Rightarrow \frac{2e^{j\omega}}{1-j(\omega-2)} \xrightarrow{FT3} e^{j2t} \cdot Z_3(t) = 2e^{j2t} \cdot e^{t-1} \cdot u(-t-1) \Leftarrow x_3(t)$

στ) $X_6(\omega) = \frac{2e^{j\omega} \cdot \sin(\omega - \pi/4)}{(\omega - \pi/4)}$ ← 2.1.1/13

$x_6(t) = e^{j\omega} \cdot e^{j\pi/4 t} \cdot [u(t+1) - u(t-1)]$

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η) $x8(\omega) = \underbrace{\frac{2}{4+j(\omega-1)}}_{FT3} \cdot \underbrace{\frac{2e^{j\omega}\cdot\sin\omega}{\pi\omega}}_{FT7}$

$\frac{2}{4+j\omega} \xrightarrow{FT1} 2e^{-4t}\cdot u(t) \longleftarrow z(t)$

$\frac{2e^{j\omega}\cdot\sin\omega}{\pi\omega} \xrightarrow{FT1} 2e^{j\omega}\cdot\frac{1}{\pi}\left[u(t+1)-u(t-1)\right]$ (2)

$\Rightarrow \frac{2}{4+j(\omega-1)} \xrightarrow{FT3} e^{jt}z(t) = 2e^{jt}\cdot e^{-4t}\cdot u(t)$ (1)

$\Rightarrow x8(t) = \frac{2}{4+j(\omega-1)}\cdot\frac{2e^{j\omega}\cdot\sin\omega}{\pi\omega} \xrightarrow{FT7} (1)\cdot(2)$

(9) $x(t) = t\left(\frac{2\sin(2t)}{\pi t}\right)^2 = t\cdot\frac{2\sin(2t)}{\pi t}\cdot\frac{2\sin(2t)}{\pi t}$

(FT11 ↑, FT8 ↗)

$\Rightarrow t\cdot\frac{2\sin(2t)}{\pi t} \xrightarrow{FT11} j\frac{dx(\omega)}{d\omega}$

$\Rightarrow t\cdot\frac{2\sin(2t)}{\pi t}\cdot\frac{2\sin(2t)}{\pi t} \xrightarrow{FT8} \frac{1}{2\pi}y(\omega)\cdot y(\omega)$

Ασκήσεις 2.3 (5/5)

(2) $H(\omega) = \frac{1}{2+2j\omega}$, $x(t) = ?$ $y(t) = e^{-2t}u(t) - e^{-5t}u(t)$

$y(\omega) = H(\omega)x(\omega)$ (LTI συστημα)

$\Rightarrow x(\omega) = \frac{y(\omega)}{H(\omega)}$

$y(\omega) = \frac{1}{2+j\omega} - \frac{1}{5+j\omega}$

$\Rightarrow x(\omega) = \frac{\frac{1}{2+j\omega} - \frac{1}{5+j\omega}}{\frac{1}{2+2j\omega}} = \frac{2+2j\omega}{2+j\omega-5+j\omega} = \frac{2+2j\omega}{2j\omega-3}$

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(3) β) $h(t) = \frac{\sin(2(t-2))}{\pi(t-2)}$

$x(t) = \frac{\sin(2(t+2))}{\pi(t+2)}$

$y(\omega) = H(\omega) \cdot X(\omega)$

$\frac{\sin(2t)}{\pi t} \xrightarrow{FT} \frac{1}{\pi} \cdot \pi[u(\omega+2) - u(\omega-2)] \leftarrow Z\omega$

$\Rightarrow \frac{\sin(2(t-2))}{\pi(t-2)} \xrightarrow{FT2} e^{-j2\omega} Z(\omega) = e^{-j2\omega}[u(\omega+2) - u(\omega-2)]$ (1)

$H(\omega) \Rightarrow$

$\Rightarrow \frac{\sin(2(t+2))}{\pi(t+2)} \xrightarrow{FT2} e^{j2\omega} Z(\omega) = e^{j2\omega}[u(\omega+2) - u(\omega+2)]$ (2)

$X(\omega) \Rightarrow$

$\Rightarrow y(\omega) = H(\omega) \cdot X(\omega) = (1) \cdot (2) \Rightarrow y(t) = F^{-1}\{y(\omega)\}$

(5) $\frac{dy(t)}{dt} + 3y(t) = 2x(t)$

α) $F\{\frac{dy(t)}{dt}\} + 3F\{y(t)\} = F\{x(t)\}$

$j\omega y(\omega) + 3y(\omega) = x(\omega)$ (FT9)

$H(\omega) = \frac{y(\omega)}{x(\omega)} = \frac{1}{3+j\omega}$

$h(t) = F^{-1}\{H(\omega)\} = e^{-3t} \cdot u \cdot t$

β) $y(\omega) = H(\omega) \cdot X(\omega)$

$= \frac{1}{3+j\omega} \cdot \frac{1}{3+j\omega} = \frac{1}{(3+j\omega)(3+j\omega)} = \frac{1/2}{(3+j\omega)} - \frac{1/2}{(3+j\omega)}$

$\Rightarrow y(t) = F^{-1}\{y(\omega)\} = y_2 \cdot e^{-3t} \cdot u(t) - \frac{1}{2} e^{-3t} \cdot u(t)$

ΠΡΩΤΟΠΟΡΙΑ ΤΗΝ ΠΡΩΤΟΥΣΑ ΣΤΟΝ ΚΟΣΜΟ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

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Ασκήσεις 4.1 (4/4) Ιωάννης Γιαγιώτης (22983)

(1) ΧΣ

$$X(s) = \mathcal{L}\{u(t)\} = \int_0^{\infty} e^{-st}\,dt = \lim_{L\to+\infty}\left[-\frac{1}{s}e^{-st}\right]_0^L$$

$$= \lim_{L\to+\infty}\left(-\frac{1}{s}e^{-Ls} + \frac{1}{s}\right) = \lim_{L\to+\infty}\left(-\frac{1}{s}\cdot e^{-L\sigma}\cdot e^{-jL\omega} + \frac{1}{s}\right)$$

$$= \frac{1}{s} \quad \text{για } Re\{s\}\ (=\sigma) > 0 \ (\pi.\chi.)$$

(2) $$\mathcal{L}\{e^{2t}u(-t)\} = \int_{-\infty}^{+\infty} e^{2t}u(-t)e^{-st}\,dt = \int_{-\infty}^{0} e^{2t}e^{-st}\,dt$$

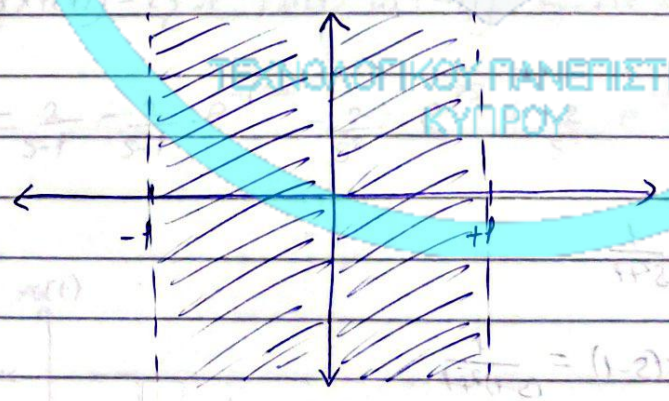
$$= \mathcal{L}\{e^{-2t}u(t)\} = \int_{-\infty}^{+\infty} e^{-2t}u(t)\cdot e^{-st}\,dt = \int_0^{+\infty} e^{-2t}\cdot e^{-st}\,dt$$

$$= \lim_{L\to+\infty}\left[e^{-2t}\cdot\left(-\frac{1}{s}e^{-st}\right)\right]_0^L = \lim_{L\to+\infty}\left[e^{-2t}\left(-\frac{1}{s}\cdot e^{-Ls} + \frac{1}{s}\right)\right]$$

$$= \lim_{L\to+\infty}\left[e^{-2t}\left(-\frac{1}{s}\cdot e^{L\sigma}\cdot e^{-jL\omega} + \frac{1}{s}\right)\right] = \frac{1}{s+2}$$

$$\Rightarrow \int_{-\infty}^{0} e^{2t}\cdot e^{-st}\,dt = \lim_{L\to+\infty}\left[\frac{1}{s+1} - \frac{1}{s-1} + \frac{1}{s-1}\cdot e^{L(s-1)}\right]_L^0 = \frac{1}{s-2}$$

για $-s < Re\{s\} < 1$



ΠΡΟΤΟΤΥΠΑ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

Watermarkly

Ασκήσεις 4.2 (9/9)

(1) a) $(5t^2-3t+2)u(t)$

$x(s)$

$= \underbrace{L\{5t^2u(t)\}}_{4.1.11\,5} - \underbrace{L\{3tu(t)\}}_{4.1.11\,5} + \underbrace{L\{2u(t)\}}_{4.1.11\,3}$ $(LT1) \rightarrow (\Pi Z \geq R_1 \cap R_2)$

$= \frac{5(2!)}{s^3} - \frac{3}{s^2} + \frac{2}{s}$ $\{Re(s)>0\} \cap \{Re(s)>-1\} \Rightarrow Re\{s\}>0$

ε) $2e^{3t}\sin(4t)u(t)$

$x(s) = L\{\sin(t)u(t)\} = \frac{1}{s^2+1}$

$\sin(4t)u(t) \xrightarrow{Lt} \frac{4}{s^2+16}$ $Z1(s)$ $(\Pi Z.\ Re\{s\}>0)$

$\Rightarrow 2e^{3t}\sin(4t)u(t) \xrightarrow{Lts} Z1(s-16) = \frac{4}{(s-16)^2+4}$ $(\Pi Z.\ Re\{s\}>4)$

$(\Pi Z. \geq \{Re\{s\}>16\} \cap \{Re(s)>0\} \Rightarrow Re\{s\}>16$

$= \frac{1}{s-7} - \frac{1}{s+1}$

ΠΡΩΤΟΠΟΡΙΑ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ ΟΥΚ ΕΛΑΤΤΩ

10a) $[\sin(2t)u(t)]''$

$x(s) = L\{\sin(t)u(t)\} = \frac{1}{s^2+1}$

$y(s) = L\{\sin(t)u(t)\} = x(s-1) = \frac{1}{(s-1)^2+1}$

$z(s) = L\{[\sin(t)u(t)]''\} = sy(s) - sy(0) - y'(0)$

$= s\cdot\frac{1}{(s-1)^2+1} - s(\sin 0) - (\sin(0t)+\cos 0) = s^2\cdot\frac{2}{s^2+4} - 2$

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(ΟΣ) $\int_{-\infty}^{t} e^{t} \cdot t\cos(t)u(t)\,dt$

$\Rightarrow \cos(t)u(t) \xrightarrow{LT} \frac{s}{s^2+1}$

$\Rightarrow t\cos(t)u(t) \xrightarrow{LT9} (-1)\frac{dX(s)}{ds} = \frac{-1}{(s-1)^2+1}$

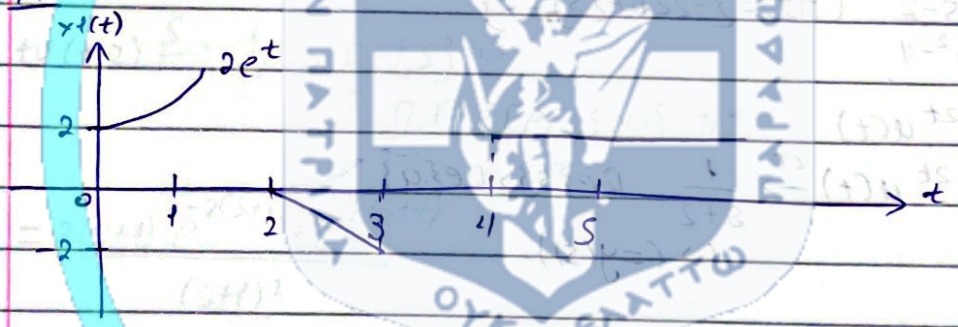
$\Rightarrow e^{t} \cdot t\cdot\cos(t)\cdot u(t) \xrightarrow{LT3} \frac{e^{-2(s-1)}}{(s+1)^2+1}$

$\Rightarrow \int_{-\infty}^{t} e^{t}\cdot t\cdot\cos(t)\cdot u(t) \xrightarrow{LT8} \frac{1}{s}X(s)$

$= \frac{1}{s}\cdot\frac{e^{-2(s+1)}}{(s+1)^2+1}$ (ΠΣ. ΙΒ)

(3)

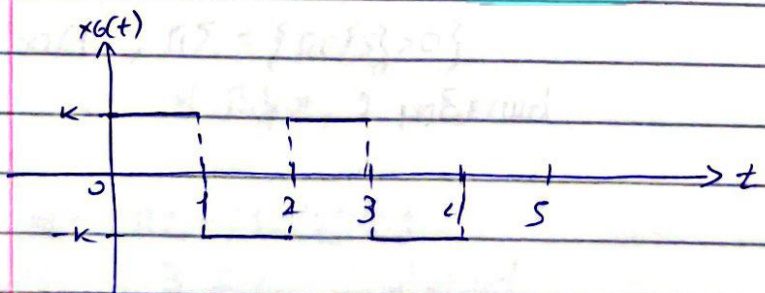
x1



$L\{x1(t)\} = L\{2e^{t}[u(t)-u(t-1)]\} + L\{-2(t-2)[u(t-2)-u(t-3)]\} + L\{2u(t-4)\}$

$= \frac{2}{s-1} - \frac{2}{s-1}\cdot e^{-(s-1)} - \frac{2}{s^2}\cdot e^{-2s} - \frac{2}{s^2}\cdot e^{-3s} + \frac{2}{s}\cdot e^{-3s} + \frac{2}{s}\cdot e^{-4s}$

x6



$x6(t) = [u(t)-u(t-1)] - [u(t-1)-u(t-2)] + [u(t-2)-u(t-3)] - [u(t-3)-u(t-4)]$

$= \frac{1}{s} - \frac{1}{s}\cdot e^{-s} - \frac{1}{s}\cdot e^{-s} - \frac{1}{s}\cdot e^{-2s} + \frac{1}{s}\cdot e^{-2s} - \frac{1}{s}\cdot e^{-3s} - \frac{1}{s}\cdot e^{-3s} - \frac{1}{s}\cdot e^{-4s}$

$= \frac{1}{s} - \frac{1}{s}\cdot e^{-s} - \frac{1}{s}\cdot e^{-s} - \frac{1}{s}\cdot e^{-3s} - \frac{1}{s}\cdot e^{-3s} - \frac{1}{s}\cdot e^{-4s}$

ΠΡΩΤΟΠΟΡΙΑ ΤΜΗΜΑ ΠΛΗΡΟΦΟΡΙΚΗΣ ΕΘΝΙΚΟΥ ΚΑΙ ΚΑΠΟΔΙΣΤΡΙΑΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΑΘΗΝΩΝ
ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ
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(4) γ) $L\{\underbrace{e^{3x}}_{LT3} \cdot \underbrace{x(t-2)}_{LT2}\}$

$x(t) \xrightarrow{LT} x(s) \Rightarrow x(t-2) \xrightarrow{LT2} e^{-2s} \cdot x(s) \; (= x1(s))$

$e^{3t} \cdot x(t-2) \xrightarrow{LT3} x(s-3) \, x1(s)$

$= x(s-3) e^{-2s} x(s)$

ε) $L\{\underbrace{[te^{-3t}}_{LT9} \cdot \underbrace{x(t)]'}_{LT3}\}$

$x(t) \xrightarrow{LT} x(s) \Rightarrow e^{-3t} x(t) \xrightarrow{LT3} x(s+3) \, Z(s)$

$\Rightarrow te^{-3t} x(t) \xrightarrow{LT9} -(Z(s))' = -x'(s+3) \; (= x2(s))$

(6) $y(t) = x(t) + a x(-t)$

LT $y(s) = \frac{s-2}{s^2-4}$ $(\Pi Z = \{-2 < Re\{s\} < 2\}$

$x(t) = be^{-2t} u(t)$

$x(t) = be^{-2t} u(t) \xrightarrow{LT} \frac{1}{s+2}$ $Re\{s\} > Re\{a\}$

$\hookrightarrow (= y(s))$

$a x(-t) \xrightarrow{LTS} \frac{1}{|a|} \cdot x\left(\frac{s}{a}\right) = \frac{1}{|2|} \cdot x\left(\frac{s}{2}\right) \; (Z_2(s))$

$y(s) = x(s) + Z(s) \Rightarrow \frac{1}{s+2} + \frac{1}{2} x\left(\frac{s}{2}\right)$

$b=1$, $a=2$

ΠΡΩΤΟΠΟΡΙΑ ΠΑΡΑΔΟΣΗ ΟΥΚ ΕΛΑΤΤΩ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΡΗΤΗΣ

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(7)

2ο

$e^{-t}(t-2)u(t-5)$ ΡΙ

ΣΕ ΠΣ. α

$s-2 \in$ ΡΙ

$(t-2) \xrightarrow{\text{ΔΤΣ}} e^{-2s}x(s)$

$e^{-t}(t-2)u(t-5) \xrightarrow{\text{ΔΤΣ}} x(s-2)$

$= \frac{1}{(s+1)^2} \cdot e^{-5(s+1)} + \frac{3}{s+1} \cdot e^{-5(s+1)}$

2η

$tu(t-2) \cdot e^{-t}$

$u(t-2) \xrightarrow{4.1.1\text{γ}} \frac{1}{s} e^{-2s}$, $Re\{s\}>0$

$tu(t-2) \xrightarrow{s} \frac{1}{s^2}$, $Re\{s\}>0$

$\overset{?}{=}$ ΡΠ $\{Re\{s\}>0\}$ ΠΣ.

$= \frac{2(s+1) \cdot e^{-2(s+1)} + e^{-2(s+1)}}{(s+1)^2}$

3

$x_1(t)$, ΠΣ. $= \{Re\{s\}>0\}$

3 πόλοι , 2 μηδενικά

$x_2(t)$, ΠΣ. $= \{Re\{s\}>0\}$

4 πόλοι , 2 μηδενικά

$x_3(t)$, ΠΣ. $= \{Re\{s\}>0\}$

3 πόλοι , 2 μηδενικά

$x_4(t)$, ΠΣ. $= \{Re\{s\}>0\}$

2 πόλοι , 3 μηδενικά

$x_5(t)$, ΠΣ. $= \{Re\{s\}>0\}$

3 πόλοι , 2 μηδενικά

$x_6(t)$, ΠΣ. $= \{Re\{s\}>0\}$

4 πόλοι , 2 μηδενικά

ΠΡΩΤΟΠΟΡΙΑ ΠΑΡΑΔΟΣΗ ΟΥΚ ΕΛΑΤΤΩ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

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Ασκήσεις 4.3

Ιωνήρος Γιαγκίζης
(22983)

(1) α) $\frac{s}{s^2+4} \overset{2'}{=} \cos 2t \cdot u(t)$

β) $\frac{1}{s^2+4} \overset{2'}{=} \sin 2t \cdot u(t)$

γ) $\frac{1}{(s-3)^2+4} \overset{2'}{=} e^{3t} \cdot \sin(2t) \cdot u(t)$

δ) $\frac{s}{(s-3)^2+4} = \frac{s-3}{(s-3)^2+4} + \frac{3}{2} \cdot \frac{2}{(s-3)^2+2^2} \overset{2'}{\Rightarrow} e^{3t}\cos 2t \cdot u(t) + \frac{3}{2} e^{3t} \sin 2t \cdot u(t)$

ε) $\frac{1}{2} \cdot \frac{2e^{-4s}}{s^2+4} \overset{2'}{=} \frac{1}{2} \sin[2(t-4)] \cdot u(t-4)$

στ) $\frac{e^{-4s} \cdot (s-3)}{(s-3)^2+4} \overset{2'}{=} \frac{1}{2} L^{-1}\left\{\frac{2(s-3)}{(s-3)^2+2^2}\right\}\Big|_{t \to t-4} = \frac{1}{2}\left[e^{3t}\sin 2t \cdot u(t)\right]_{t-4}$

$= \left[\frac{1}{2} e^{3t} \sin(2t-4)\right] \cdot u(t-4)$

ζ) $\frac{1}{9s^2+4} = \frac{1}{9(s^2+\frac{4}{9})} = \frac{1}{9} \cdot \frac{1}{s^2+\left(\frac{2}{3}\right)^2} = \frac{1 \cdot 3}{9 \cdot 2} \cdot \frac{2/3}{s^2+\left(\frac{2}{3}\right)^2} \overset{2^{-1}}{\Rightarrow} \frac{1}{6} \sin\left(\frac{2}{3}t\right)$

η) $\frac{1}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1}$ $\begin{matrix} s=0: A=1/2 \\ s=1: B=-1/2 \\ s=-1: C=1/2 \end{matrix}$ $\Rightarrow \frac{1}{2s} - \frac{1}{2(s^2+1)}$ *

θ) $\frac{e^{-2(s-3)}}{(s-3)^2+1} \Rightarrow L^{-1}\left\{\frac{e^{-2(s-3)}}{(s-3)^2+1}\right\} = e^{3t} L^{-1}\left\{\frac{e^{-2s}}{s^2+1}\right\} = e^{3t}\left[L^{-1}\left\{\frac{1}{s^2+1}\right\}\right]_{t \to t-2}$

$= e^{3t}[\sin t \cdot u(t)]_{t \to t-2} = e^{3t} \cdot \sin(t-2) \cdot u(t-2)$

* $\Rightarrow 2\left\{\frac{1}{2s}\right\} = \frac{1}{2} \cdot L\left\{\frac{1}{s}\right\} = \frac{1}{2} \cdot \frac{1}{t} = \frac{1}{2t} \Rightarrow 2\left\{\frac{1}{2(s^2+1)}\right\} = -\frac{1}{2} \cdot 2\left\{\frac{1}{s^2+1}\right\} = -\frac{1}{2} \cdot \frac{1}{2} \cdot 2\left\{\frac{\frac{1}{2}}{s^2+1}\right\}$

$= -\frac{1}{4} \cdot 2\left\{\frac{\frac{1}{2}}{s^2+1}\right\}$ $L \Rightarrow \left\{\frac{1}{\left(\frac{s^2}{2}+1\right)}\right\} = \sqrt{2} \cdot L\left\{\frac{1}{\sqrt{2}} \cdot \frac{\frac{\sqrt{2}}{2}}{s^2+1}\right\} = \sqrt{2} \cdot \sin\left(\frac{\sqrt{2}}{2}t\right) \Rightarrow \frac{1}{2t} - \frac{1}{4}\sqrt{2}\sin\left(\frac{\sqrt{2}}{2t}\right)$

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(6) α) $\mathcal{L}^{-1}\{x''(s)\} = \mathcal{L}^{-1}\left\{(-1)^2 \frac{d^2 X(s)}{ds^2}\right\} = \mathcal{L}^{-1}\{\mathcal{L}\{t^2 x(t)\}\} = t^2 \cdot x(t)$

β) $\mathcal{L}^{-1}\left\{\frac{X(s-3)}{s-3}\right\} = e^{3t}\mathcal{L}^{-1}\left\{\frac{X(s)}{s}\right\} = e^{3t}\int_{-\infty}^{t} x(\tau)d\tau$

γ) $\mathcal{L}^{-1}\left\{\frac{X(s+4)}{s}\right\} = \int_{-\infty}^{t} \mathcal{L}^{-1}\{X(s+4)\}dt = \int_{-\infty}^{t} e^{-4t}x(t)\,dt$

δ) $\mathcal{L}^{-1}\{e^{-3(s-1)}X(s-1)\} = e^{t}\mathcal{L}^{-1}\{e^{-3s}\cdot X(s)\} = e^{t}\cdot x(t-3)$

(7) α) $\frac{1}{s+2} + \frac{1}{s-2} \xrightarrow{-12} [e^{-2t} + e^{2t}]u(t)$

$\frac{\int -2 + \int +2}{(s+2)\cdot(s-2)} = \frac{2s}{(s+2)(s-2)} = \frac{2s}{s^2-4} = \lim_{s\to\infty}\frac{2s}{s^2} = \frac{2}{s} = 0$

πόλοι: $s = \pm 2$

Μηδενικό στο 0 και ∞



β) $\frac{s-2}{s^2-4} = \frac{s-2}{(s-2)(s+2)} = \frac{1}{s+2}$

πόλος: $s = -2$

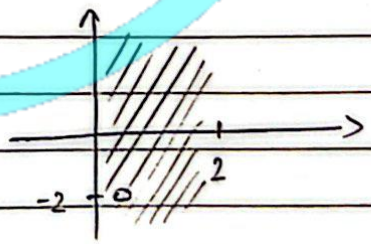
Μηδενικό στο ∞



γ) $\frac{s^2-4}{s^2-4s+4} = \frac{(s-2)(s+2)}{(s-2)^2} = \frac{s+2}{s-2}$

πόλος: $s = 2$

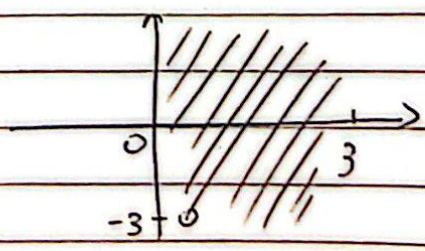
Μηδενικό: $s = -2$



δ) $\frac{s^3-9s}{s^2-6s+9} = \frac{s(s^2-9)}{(s-3)^2} = \frac{s(s-3)(s+3)}{(s-3)^2} = \frac{s(s+3)}{s-3}$

πόλος: $s = 3$

Μηδενικό: $s = -3, s = 0$



ΠΡΩΤΟΠΟΡΙΑ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ
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(8) α) $\frac{s-1}{(s+1)(s^2+6s+9)} = \frac{A}{s+1} + \frac{B}{s+3} + \frac{C}{(s+3)^2}$

$s-1 = A(s+3)^2 + B(s+1)(s+3) + C(s+1)$

$s=-1: -2 = 2A - 1B - 2C$

$s=-3: C = -6$

$4 = -6A - 2B$

$s=-1: 2 = 4A + 2B$

$\Rightarrow A = \frac{1}{4}, B = -\frac{3}{4}, C = -6$

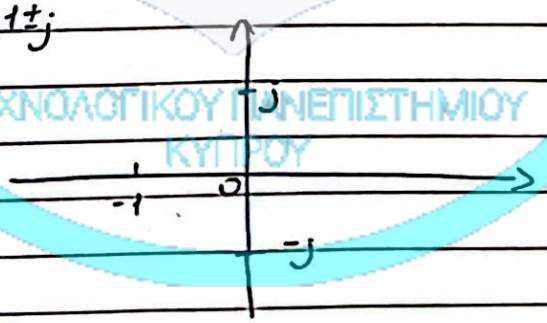
$\frac{s-1}{(s+1)(s^2+6s+9)} = \frac{1/4}{s+1} - \frac{3/4}{s+3} - \frac{6}{(s+3)^2}$

$L^{-1} \Rightarrow \frac{1}{4}e^{-t} - \frac{3}{4}e^{-3t} - 6e^{-3t}t$

β) $\frac{2}{s(s+1)(s^2+2s+2)}$

$\Rightarrow \frac{-B \pm \sqrt{B^2-4\alpha\gamma}}{2a} = \frac{-2 \pm \sqrt{2^2-4\cdot1\cdot2}}{2\cdot1} = \frac{-2 \pm 2j}{2} = -1 \pm j$

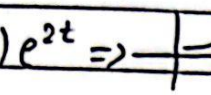
Πόλοι: $-1, 0, -1 \pm j$



$\Rightarrow$ Ζώσημα //

ΠΡΩΤΟΠΟΡΙΑ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ
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Ασκήσεις 4.4

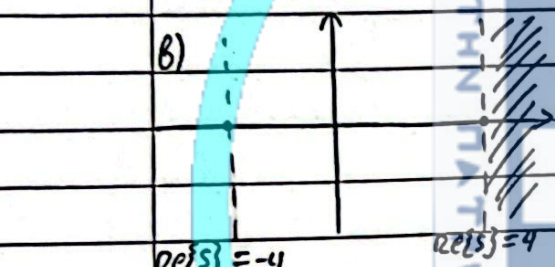
(1) α) $e^{2t} \Rightarrow$  , αιτιατό

β) $H(s) = \frac{1}{s-2}$, Π.Σ: $Re\{s\} < 2$

γ) Πόλος: $s = 2 \Rightarrow$ όχι ευσταθές

(3) $H(s) = \frac{1}{(s-4)(s+4)} = \frac{A}{s-4} + \frac{B}{s+4}$

α) $Re\{s\} > -4$
$Re\{s\} - Re\{-4\}$ } Π.Σ = $Re\{s\} > 4$

β) 

$\Rightarrow \{s\} > 4$, αιτιατό

γ) $\frac{118}{s-4} - \frac{118}{s+4} \overset{\mathcal{L}^{-1}}{=} \left[\frac{1}{8}e^{4t} - \frac{1}{8}e^{-4t}\right] \cdot u(t)$

δ) όχι ευσταθές

ΠΡΩΤΟΠΟΡΙΑ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

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(6) $x(t) = e^{-t} \cdot u(t)$

$h(t) = e^{-2t} \cdot u(t)$

α) $X(s) = \frac{1}{s+1}$, $H(s) = \frac{1}{s+2}$

β) $Y(s) = X(s) \cdot H(s)$

$= \frac{1}{s+1} \cdot \frac{1}{s+2} = \frac{1}{(s+1)(s+2)}$

$Y(s) = \frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \Rightarrow 1 = A(s+2) + B(s+1)$

$s = -2: \; B = 1/2$

$s = -1: \; A = -1/2$

$\Rightarrow Y(s) = \frac{-1/2}{s+1} + \frac{1/2}{s+2} \xrightarrow{L^{-1}} \left[\frac{-1}{2} e^{-t} + \frac{1}{2} e^{-2t} \right] u(t)$

γ) $y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) \, d\tau$

$= \int_{-\infty}^{t} e^{-t} u(\tau) e^{-2(t-\tau)} u(t-\tau) \, d\tau$

$= \int_{0}^{t} e^{-t} e^{-2t} e^{2\tau} \, d\tau = \int_{0}^{t} e^{-2\tau} e^{-2t}$

$= \left[\frac{-1}{2} e^{-2\tau} e^{-2t} \right]_{0}^{t} = \frac{-1}{2} e^{-t} + \frac{1}{2} e^{-2t}$

$= \frac{1}{2} e^{-2t} - \frac{1}{2} e^{-t}$ για $t \geq 0$.

(5)

ΠΡΩΤΟΠΟΡΙΑ ΠΑΡΑΔΟΣΗ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ
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